<u>TUTORIAL THREE</u>

CPU Scheduling

1. State whether each of the following statements are true or false. Justify your answers.

   (a) When a new process is admitted in the system, the short-term scheduler must execute in order to keep the CPU busy.

   (b) Partitioned multi-processor scheduling suffers from migration overheads due to data in private core-specific caches.

2. Consider the following set of processes that are using the same I/O device.

Process	CPU Burst Times	I/O Burst Time	Arrival Time
P1	2,8	2 (after first CPU burst)	0 (order: 1)
P2	1	0	0 (order: 2)
P3	1,1	1 (after first CPU burst)	2 (order: 1)
P4	1	0	2 (order: 2)
P5	2,3	2 (after first CPU burst)	4

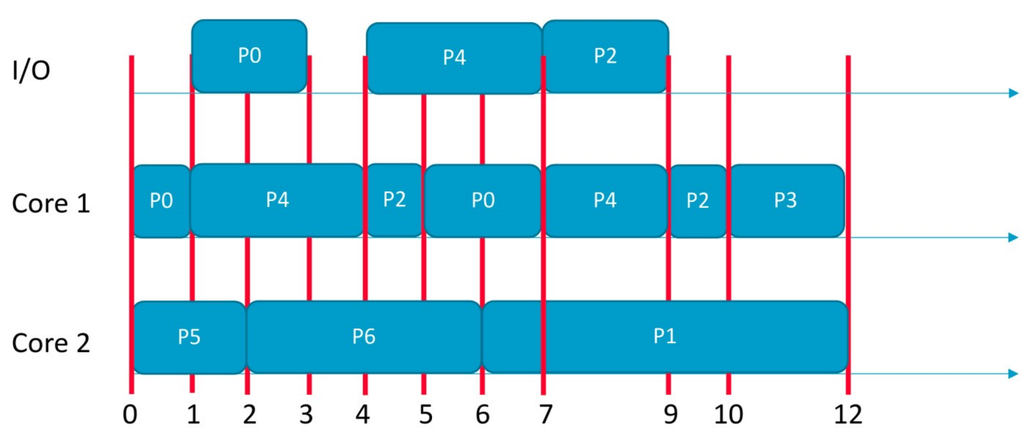
   (a) Draw two Gantt charts illustrating the execution of these processes using:
      i. Round-Robin (quantum=2) uni-processor scheduling.
      ii. Shortest Remaining Time First (SRTF) global multi-processor scheduling with 2 cores.

   (b) What are the turnaround and waiting times of processes for each algorithm?

3. Consider a set of processes P0-P6 with the following arrival times.

Process	Arrival Time Instant
P0, P1, P4, P5, P6	0 (order: P0, P1, P4, P5, P6)
P2	3
P3	9

A dual-core schedule of these processes is shown in the figure below.



(a) What are the CPU and I/O burst lengths for each process (if there are multiple CPU or I/O bursts for a process, then identify all of them).

(b) Does the schedule represent partitioned or global multiprocessor scheduling? Identify the scheduling algorithm(s) used in the schedule (select from those covered in the lectures). Justify your answer.

4. Measurements of a certain system have shown that the average process runs for time T before blocking on I/O. A process switch requires time S, which is effectively wasted (overhead). Define what is meant by CPU efficiency. For round robin scheduling with quantum Q, give a formula for the CPU efficiency for each of the following cases:

(a) $Q = \infty$

(b) $Q > T$

(c) $S < Q < T$

(d) $Q = S$

(e) $Q = 0$